

Exercise 5.2: Detection Pipeline

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Dataset

We used the provided Roboflow balloon detection dataset in COCO format, split into 42 training, 12 validation, and 6 test images (416×416 px, CC BY 4.0). No augmentation was applied at the dataset level; augmentation was introduced only inside our pipeline (see Q5.2.3).

Pipeline Overview

Our pipeline follows five stages:

1. **Region proposals.** Selective Search is run at three Felzenszwalb scales ($k \in \{100, 150, 250\}$) and the resulting proposals are pooled, deduplicated, and filtered by minimum area and aspect ratio. Proposals are cached to disk so the slow proposal step runs only once.
2. **Sample labelling.** Each proposal is compared with every ground-truth box via IoU. Proposals with $\text{IoU} \geq t_p$ are positive (balloon); those with $\text{IoU} \leq t_n$ are negative (background); those in the middle band (t_n, t_p) are treated as *hard negatives* and included in training with a negative label.
3. **Feature extraction.** Each proposal crop is resized to 64×64 and described by a concatenation of (i) HOG (9 orientations, 8×8 cells, 2×2 blocks, L2-Hys normalisation; 1764-dim), (ii) HSV colour histogram (16 bins per channel; 48-dim), and (iii) two shape features (aspect ratio and a circularity proxy; 2-dim), giving a 1814-dim descriptor. Positive crops are augmented with horizontal flips and ± 0.1 brightness offsets, tripling the number of positive training examples.
4. **SVM training.** A `LinearSVC` with `class_weight='balanced'` is trained inside a `StandardScaler` pipeline. The regularisation constant C is selected by 3-fold cross-validation on the training set over $C \in \{0.01, 0.1, 1.0, 10.0\}$, maximising balloon F1.
5. **Inference and evaluation.** At test time, all proposals scoring above a threshold are passed through Non-Maximum Suppression (NMS). Predictions are saved as COCO-style JSON and evaluated with the `pycocotools` API (AP@0.50 and mAP@0.50:0.95). Proposal quality is separately measured with MABO (Mean Average Best Overlap) following Uijlings et al.

Experiments

We ran seven experiments varying the score threshold, NMS threshold, and aspect-ratio filter. All experiments share the same multi-scale proposal set ($k \in \{100, 150, 250\}$, max aspect ratio unless noted) and the same SVM trained with $t_p = 0.5$, $t_n = 0.25$. MABO and the SVM were identical across experiments 1–5 because the proposals and training data were cached.

Table 1: Summary of experiment results on the test set. “Thresh” is the SVM decision score threshold; “Asp.” is the maximum allowed proposal aspect ratio. Best AP@0.50 in **bold**.

Exp.	Description	Thresh	NMS	Asp.	AP@0.50	mAP@0.50:0.95
01	Balanced	0.30	0.30	2.0	0.301	0.095
02	High recall	0.10	0.50	2.0	0.298	0.107
03	Clean/precise	0.50	0.30	2.0	0.301	0.095
04	Strict	0.70	0.30	2.0	0.256	0.085
05	Auto threshold	0.076	0.30	2.0	0.301	0.095
06	Wider aspect	0.30	0.30	3.0	0.241	0.090
07	Narrow aspect	0.30	0.30	1.5	0.144	0.058

Proposal quality (MABO)

MABO measures the average best-overlap between each ground-truth balloon box and the closest proposal, independently of the classifier. Across all experiments that used the same proposals (Exp. 01–05), the scores were:

Split	Train	Validation	Test
MABO	0.676	0.619	0.622

A test MABO of 0.622 indicates that on average the best available proposal covers about 62% of each balloon’s area. This is a meaningful but imperfect upper bound: the classifier can only detect what the proposals cover, so roughly 38% of balloon area is structurally unrecoverable at this configuration. Widening the aspect-ratio filter (Exp. 06, aspect ≤ 3.0) did not improve MABO (test: 0.617) despite generating roughly 40% more proposals, which shows that additional elongated proposals do not help cover the approximately-circular balloon shape better. Conversely, restricting to nearly-square proposals (Exp. 07, aspect ≤ 1.5) produced the highest test MABO (0.641) but also the fewest proposals, and the downstream AP dropped sharply to 0.144 because too many valid proposals were discarded before the SVM had a chance to score them.

SVM selection

The cross-validation consistently preferred $C = 0.01$ (Exp. 01–06), indicating that strong regularisation is beneficial when the training set is small. Exp. 07 selected $C = 1.0$ because the narrower proposal filter changed the class distributions enough to shift the optimum.

C	0.01	0.1	1.0	10.0
Val. F1 (Exp. 01)	0.366	0.280	0.269	0.267

Effect of score threshold

Experiments 01–05 used the same SVM but varied the score threshold. Table 2 shows the precision–recall trade-off on the validation set.

Lowering the threshold (Exp. 02) raises recall at the cost of precision. The AP@0.50 values are nearly identical for Exp. 01, 02, 03, and 05 (0.298–0.301), confirming that threshold choice mainly shifts the precision–recall operating point without changing the area under the curve. The strict threshold (Exp. 04, 0.70) degrades AP to 0.256, suggesting that the SVM confidence scores are not well-calibrated enough to sustain high precision at a very high cut-off.

Table 2: Validation classification report at different thresholds (Exp. 01–04). Support: 1105 background, 83 balloon.

Exp.	Thresh	Bal. Prec.	Bal. Recall	Bal. F1	Accuracy
01	0.30	0.30	0.34	0.32	0.90
02	0.10	0.30	0.42	0.35	0.89
03	0.50	0.32	0.29	0.30	0.91
04	0.70	0.34	0.24	0.28	0.91

Best configuration

Experiments 01, 03, and 05 achieve the highest AP@0.50 of 0.301. We report Exp. 01 as the primary result because its parameters (score threshold 0.30, NMS 0.30, aspect ratio 2.0) were set without using the test labels. Exp. 02 achieves the highest mAP@0.50:0.95 of 0.107 by trading precision for recall across the full IoU range, which is preferable when localisation accuracy matters.

Questions

Q5.2.1 — Differences from Uijlings et al.

Our implementation simplifies the original pipeline in several ways. Uijlings et al. extract SIFT descriptors and encode them into a bag-of-words or Fisher Vector representation that captures richer local structure than our HOG + colour histogram. They train *one SVM per object category* and use the full PASCAL VOC training set, whereas we train a single binary SVM on 42 images. Their paper also evaluates multiple complementary grouping strategies in Selective Search simultaneously (different colour spaces and similarity combinations), while our implementation runs a single colour-space strategy. Finally, they apply bounding-box regression to refine localisation after classification, a step we omit entirely. The practical consequence is visible in the results: our AP@0.50 of 0.301 is well below what region-proposal methods achieved on PASCAL when paired with stronger features.

Q5.2.2 — Effect of the two thresholds

The positive threshold t_p controls how tightly a proposal must overlap a ground-truth box to be used as a balloon example. Raising t_p (e.g. to 0.75) yields fewer but higher-quality positive samples; lowering it (e.g. to 0.50, as we used) produces more positives at the cost of some label noise, because a proposal at IoU 0.50 may include as much background as balloon.

The negative threshold t_n controls the lower cut-off for background examples. Two thresholds are needed rather than one because proposals in the middle band (t_n, t_p) are genuinely ambiguous: they partially cover a balloon but not enough to be a reliable positive. Including them as positives would introduce label noise; including them as negatives would train the SVM to suppress partially-visible balloons. The gap region is therefore either discarded (standard practice) or used as *hard negatives* (our approach), which encourages the SVM to place its decision boundary where the ambiguity actually lies. In our pipeline, hard negatives contributed to a modest improvement in balloon recall compared to a version that discarded the gap entirely.

Q5.2.3 — Increasing the amount of training data

Several strategies can enlarge the effective training set without additional annotation:

- **Geometric augmentation.** Horizontal flips, small rotations, and scale jitter are label-preserving for round objects like balloons. We applied horizontal flips and ± 0.1 brightness offsets to all positive crops, tripling the number of positive training examples at no annotation cost.
- **Ground-truth boxes as proposals.** Including the annotated boxes directly as positive training examples guarantees at least one high-quality positive per balloon in every image, compensating for cases where Selective Search fails to generate a proposal with $\text{IoU} \geq t_p$.
- **Multi-scale proposals.** Running Selective Search at multiple Felzenszwalb scales and pooling the results increases proposal diversity and therefore the number of IoU-qualifying positives.
- **Hard-negative mining.** Proposals in the ambiguous IoU band (t_n, t_p) that are confidently misclassified by the SVM can be added to the negative training set in a second pass, providing additional signal without new images.
- **External data or transfer learning.** Pre-trained CNN features (e.g. from a model trained on ImageNet) would provide a richer representation that generalises better from few examples. Similarly, other balloon image datasets could be incorporated if available under a compatible licence.